

16

Cumartesi Saturday

Ekim-October

Hafta Week 42

$$\textcircled{1} \int_0^1 x^2 \left(\ln \frac{1}{x}\right)^3 dx = ?$$

$$\ln \frac{1}{x} = u \Rightarrow \frac{1}{x} = e^u \quad x = e^{-u}$$

$$-\frac{dx}{x} = du$$

$$\begin{aligned} x \rightarrow 1 & \quad u \rightarrow 0 \\ x \rightarrow 0 & \quad u \rightarrow \infty \end{aligned}$$

$$-\int_{\infty}^0 e^{-2u} u^3 \frac{-1}{e} du = \int_0^{\infty} u^3 e^{-3u} du$$

$$3u = t \quad 3 du = dt$$

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Pazar Sunday

$$\int_0^{\infty} \left(\frac{t}{3}\right)^3 \cdot e^{-t} \frac{dt}{3}$$

Ekim-October

Hafta Week 42

$$\frac{1}{81} \int_0^{\infty} t^3 \cdot e^{-t} dt = \frac{1}{81} \Gamma(4) = \frac{3!}{81} = \frac{2}{27}$$

$$n-1=3 \Rightarrow n=4$$

$$\left(\Gamma(n) = \int_0^{\infty} t^{n-1} e^{-t} dt \right)$$

Ekim / October-2010

cu
01pt
04sa
05ça
06pe
07cu
08pt
11sa
12ça
13pe
14cu
15pt
18sa
19

Kasım / November-2010

pt
01sa
02ça
03pe
04cu
05pt
08sa
09ça
10pe
11cu
12pt
15